Git Push

TOPIC Finding the Principle Components DATE

PCA

$$x = \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 3 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix} \end{array} \qquad d = 3$$

Step 1 Mean Centering

$$\mu = [1.5 \quad 1.5 \quad 1.5] \qquad x - \mu$$

$$X_c = \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{bmatrix} 0.5 & -1.5 & -0.5 \\ -1.5 & 0.5 & -0.5 \\ 1.5 & -0.5 & 0.5 \\ -0.5 & 1.5 & 0.5 \end{bmatrix} \end{array}$$

Step 2 Covariance Matrix

$Cov(x_1, x_1)$

$Cov(x_1, x_2)$

$Cov(x_1, x_3)$

$Cov(x_2, x_1)$

$Cov(x_2, x_2)$

$Cov(x_2, x_3)$

$Cov(x_3, x_1)$

$Cov(x_3, x_2)$

$Cov(x_3, x_3)$

	x_1	x_2	x_3
x_1	1.25	-0.75	0.25
x_2	-0.75	1.25	0.25
x_3	0.25	0.25	0.25

Step 3 Eigen values & Eigenvectors of Covariance matrix

$$\therefore \det(X - \lambda I) = 0$$

$$\begin{vmatrix} 1.25-\lambda & -0.75 & 0.25 \\ -0.75 & 1.25-\lambda & 0.25 \\ 0.25 & 0.25 & 0.25-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 1.25-\lambda & -0.75 & 0.25 \\ -0.75 & 1.25-\lambda & 0.25 \\ 0.25 & 0.25 & 0.25-\lambda \end{vmatrix} = 0$$

$$(1.25-\lambda)\left[(0.25-\lambda)(1.20-\lambda) - 0.20\times 0.25\right]$$

$$\lambda^3 - 2.76\lambda^2 + 1.515\lambda - 0.0025 = 0$$

$\lambda_1 = 2.186$

$\lambda_2 = 0.572$

$\lambda_3 = 0.002$

Total variance

$= 0.002 + 0.0572 + 2.186$

$= 2.763$

$PC1 = \frac{2.186}{2.763} = 79\%$

$PC2 = 20\%$

$PC3 = 0.07\%$

1D will explain 79%

2D will explain 99.9%